# MAgent Theory 2.0

Supplementary Proofs

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This supplement gives supporting calculations and secondary proofs. Each result restates its assumptions and uses the same notation as the main text. The main constructions and their decisive proofs remain in the main manuscript and its appendices.

# Chapter S1

# Probability and Predictive Reconstruction

## S1.1 The conditional relative-entropy identity

The disintegrations used here are regular conditional laws on standard Borel spaces [1].

Let $\mu, \nu$ be probability laws on a product $X \times Y$ of standard Borel spaces, with $D_{\mathrm{KL}}(\mu\|\nu) < \infty$. Write their marginals as $\mu_Y, \nu_Y$ and their conditional laws as $\mu^y, \nu^y$. Then

$$D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(\mu_Y\|\nu_Y) + \int D_{\mathrm{KL}}(\mu^y\|\nu^y)\mu_Y(dy). \tag{S1.1}$$

To see the factorization, put $a = d\mu_Y/d\nu_Y$ and $b_y = d\mu^y/d\nu^y$ for $\mu_Y$-almost every $y$. Integration on measurable rectangles shows that $a(y)b_y(x)$ is a version of $d\mu/d\nu$. The logarithm splits into $\log a + \log b_y$. Integrating these terms gives eq. (S1.1). The hypotheses guarantee the required conditional versions and prevent an undefined subtraction of infinite quantities. This argument underlies the predictive and reconstruction identities in the main text.

## S1.2 Different predictive mechanisms

Let $s, t$ be laws on a model space and let $K, L$ be normalized kernels into a common record space. Suppose the following joint divergence is finite. Define

$$\mu(dm, do) = s(dm)K(m, do), \qquad \nu(dm, do) = t(dm)L(m, do). \tag{S1.2}$$

Disintegration first by $m$ and then by $o$ gives the exact equality

$$\begin{aligned} D_{\mathrm{KL}}(s\|t) + \int D_{\mathrm{KL}}(K(m,\cdot)\|L(m,\cdot))s(dm) &= D_{\mathrm{KL}}(sK\|tL) \\ &\quad + \int D_{\mathrm{KL}}(\mu^o\|\nu^o)(sK)(do). \end{aligned} \tag{S1.3}$$

This is two uses of eq. (S1.1) on one joint law. It displays explicitly the distinction contributed by the evaluators. Equal model laws do not remove that term.

For a bijective change of model presentation $\Theta$, replace $t$ by $\Theta_{\#}t$ and replace $L$ by $L(\Theta^{-1}m, \cdot)$. Both sides then compare laws in one presentation. Measurable record relabeling must be applied consistently as well.

## S1.3 A conditional reconstruction on four points

Let $E = \{0,1\}^2$, write its coordinates as $(H, V)$, and transmit only $H$. Choose a sender law with

$$q(H = 1) = \frac{1}{2}, \qquad q(V = H \mid H) = 1. \tag{S1.4}$$

Let the receiver's prior $p$ make $H, V$ independent uniform bits. The communicated marginal is uniform, so reconstruction using $p(V \mid H)$ returns the uniform law on all four points. Its transmitted marginal is exactly correct, but it discards the sender's perfect dependence. Direct evaluation gives

$$D_{\mathrm{KL}}(q\|\widehat{q}) = 2\left(\frac{1}{2}\log\frac{1/2}{1/4}\right) = \log 2. \tag{S1.5}$$

The conditional divergence at each value of $H$ is also $\log 2$, verifying the conditional identity exactly. A receiver whose prior has the same conditional relation $V = H$ instead reconstructs $q$ from the transmitted marginal. The difference comes from its prior conditional, not from a larger message.

## S1.4 A shared source and two routes

Let $X$ be a uniform bit. Two routes return $Y_1 = X$ and $Y_2 = X \oplus b$, where $b \in \{0,1\}$ is a fixed route mark and $\oplus$ is addition modulo two. The comparison $O = Y_1 \oplus Y_2$ equals $b$ with probability one. If the second route instead uses an independent uniform bit $X'$, the same comparison is uniform and reveals no route mark. The marginal law arriving on each route is uniform in both experiments. Thus the shared-source dependence, not those marginals, determines the comparison outcome. A record model intended to inspect a loop must specify which occurrences refer to the same source.

## S1.5 Predictive sufficiency for a selected response

Let $X = (A, B)$ be two independent uniform bits, let the encoder retain $U = A$, and take no side information. For target $Y = A$, one has $I(Y; X \mid U) = 0$. For target $Y = B$, the same representation gives $I(Y; X \mid U) = \log 2$. Perfect predictive retention is therefore a statement about a specified target. If a later interaction reads $B$, the first sufficiency statement does not supply its response. This elementary distinction persists when $X$ is a complete configuration of law-valued agents and $Y$ is a boundary process.

# Chapter S2

# Algebra of Selective Exponential Links

This chapter gives supporting matrix identities for interactions written in a reference vector space. All spaces are finite dimensional and real. The results concern arbitrary finite matrices and do not require Gaussian laws, diagonalizability, or invertibility of the generator. A statistical interpretation enters only when the matrices are assigned to interfaces or made functions of law parameters.

## S2.1 The low-rank exponential

Let $O : \mathbb{R}^r \to \mathbb{R}^d$ be a receiving map and $T : \mathbb{R}^d \to \mathbb{R}^r$ a sending map. Define the entire matrix function

$$\varphi_1(Z) = \sum_{n=1}^{\infty} \frac{Z^{n-1}}{n!}. \tag{S2.1}$$

When $Z$ is invertible this equals $(e^Z - I)Z^{-1}$, but the series defines it also at singular $Z$.

**Lemma S2.1 (Exponential through the message space).** The following identity holds for all $O, T$:

$$e^{OT} = I_d + O\varphi_1(TO)T. \tag{S2.2}$$

*Proof.* Associativity gives $(OT)^n = O(TO)^{n-1}T$ for every integer $n \geq 1$. Absolute convergence of the finite-dimensional exponential series permits summation and yields eq. (S2.2). □

In particular, the correction to identity has image in $\operatorname{im} O$ and vanishes on $\ker T$. The image of the full exponential need not be low dimensional. Indeed its inverse is $e^{-OT}$, so the identity contribution preserves all input dimensions even when the communicated correction is selective.

The identity also applies to several interfaces simultaneously. For maps $O_h, T_h$, stack $O = [O_1 \ \cdots \ O_H]$ and stack the $T_h$ vertically to form $T$. Then $OT = \sum_h O_h T_h$ and the same formula holds. The smaller matrix $TO$ includes all cross-interface products $T_h O_k$; replacing it by its diagonal blocks would require additional vanishing conditions.

**Proposition S2.2 (Rank-one closed form and determinant).** For column vectors $u, v \in \mathbb{R}^d$, put $L = uv^T$ and $t = v^T u$. Then

$$e^L = I_d + \varphi_1(t)L, \qquad \det e^L = e^t. \tag{S2.3}$$

If $t = 0$, then $L^2 = 0$ and $e^L = I_d + L$.

*Proof.* The first identity is eq. (S2.2) with a one-dimensional message space. For any square matrix $A$, Jacobi's determinant identity applied to $X(s) = e^{sA}$ gives $\frac{d}{ds} \det X(s) = \operatorname{tr}(A) \det X(s)$.

Since $X(0) = I$, its determinant is $e^{s\,\mathrm{tr}(A)}$. Set $A = L$, $s = 1$, and use $\mathrm{tr}(uv^T) = v^T u$. Finally $L^2 = u(v^T u)v^T$ proves the nilpotent case. □

This result distinguishes a singular generator from a singular interaction: a rank-one generator gives an invertible exponential in every dimension. It does not assert that every invertible real matrix is one exponential.

## S2.2 Differentiating a law-dependent link

**Lemma S2.3 (Derivative of the matrix exponential).** For square matrices $L, H$, the Fréchet derivative is

$$D\exp_L[H] = \int_0^1 e^{(1-t)L} H e^{tL} dt. \tag{S2.4}$$

*Proof.* Let $X_\epsilon(t) = e^{t(L+\epsilon H)}$. Differentiating its matrix differential equation with respect to $\epsilon$ at zero gives $\dot{Y} = LY + He^{tL}$, $Y(0) = 0$. Multiplying by $e^{-tL}$ and integrating gives $Y(t) = \int_0^t e^{(t-s)L} H e^{sL} ds$. Evaluation at $t = 1$ proves the formula. Smooth dependence is also obtained directly by differentiating the locally uniformly convergent exponential series. □

If $[L, H] = 0$, this derivative reduces to $e^L H$. Without that hypothesis the replacement is invalid in general. For a link $\Theta = U_i e^L U_j^{-1}$ with fixed endpoint frames, its variation is $U_i D\exp_L[dL]U_j^{-1}$. If the frames vary, the product rule supplies their two additional terms, including $d(U_j^{-1}) = -U_j^{-1}(dU_j)U_j^{-1}$.

For the moment-dependent family $L = \lambda \mu_i \mu_j^T M_j^{-1}$, with fixed scalar $\lambda$, ordinary differentiation gives

$$\begin{aligned} dL = \lambda\big[&(d\mu_i)\mu_j^T M_j^{-1} + \mu_i (d\mu_j)^T M_j^{-1} \\ &- \mu_i \mu_j^T M_j^{-1}(dM_j)M_j^{-1}\big]. \end{aligned} \tag{S2.5}$$

Here a differentiable family of laws is assumed to have differentiable first and second moments; differentiation under their defining integrals requires the corresponding domination or integrability hypotheses. Positive-definite covariance makes inversion smooth. Combining eqs. (S2.4) and (S2.5) gives the complete link derivative on this admitted family, with no assumption about the shape of its densities.

The resulting terms enter a variational derivative whenever an interaction energy reads the link. Holding the generator fixed suppresses them and defines a different update convention. This algebra separates the constitutive choice of a law-dependent interaction from the kinetic choice used to evolve the laws; neither choice follows from the exponential parameterization alone.

# Chapter S3

# Gaussian comparison costs and repeated variables

This chapter distinguishes a Gaussian comparison likelihood from a divergence between Gaussian laws by direct calculation. The comparison likelihood is normalized in its record. Its expectation depends on the joint law of the contents that enter the comparison, including whether two appearances denote the same variable. The calculations require finite second moments; Gaussian recognition is needed only where a Gaussian relative entropy is evaluated.

## S3.1 The expected comparison likelihood

Let $X_i \in \mathbb{R}^{n_i}$ and $X_j \in \mathbb{R}^{n_j}$ have means $\mu_i, \mu_j$, covariances $\Sigma_{ii}, \Sigma_{jj}$, and cross covariance $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. Let $A : \mathbb{R}^{n_i} \to \mathbb{R}^d$ and $B : \mathbb{R}^{n_j} \to \mathbb{R}^d$ be fixed linear maps. For $R \succ 0$, define the normalized record law

$$O \mid X_i, X_j \sim \mathcal{N}(AX_i - BX_j, R), \qquad c_R = \tfrac{1}{2} \log \det(2\pi R). \tag{S3.1}$$

**Proposition S3.1 (Complete residual covariance).** At a fixed record $o$, set $\delta = A\mu_i - B\mu_j - o$. Then

$$\begin{aligned} \mathbb{E}[-\log p(o \mid X_i, X_j)] &= c_R + \tfrac{1}{2}\delta^T R^{-1}\delta + \tfrac{1}{2}\operatorname{tr}(R^{-1}V_{ij}), \\ V_{ij} &= A\Sigma_{ii}A^T + B\Sigma_{jj}B^T - A\Sigma_{ij}B^T - B\Sigma_{ji}A^T. \end{aligned} \tag{S3.2}$$

*Proof.* The negative log density is $c_R + \frac{1}{2}(AX_i - BX_j - o)^T R^{-1}(AX_i - BX_j - o)$. Write its residual as its mean $\delta$ plus a centered variable. The linear cross terms have zero expectation, and the centered quadratic has expectation $\operatorname{tr}(R^{-1}V_{ij})$. Expanding the covariance of $AX_i - BX_j$ gives the stated four terms. □

Independence is sufficient to remove $\Sigma_{ij}$, but it is stronger than necessary. Only the cross covariance visible through $A$ and $B$ contributes to this cost. A pair can be dependent while both displayed cross terms vanish. Conversely, knowing the two marginal laws does not generally determine the expected likelihood.

## S3.2 Equal rows need not give equal covariance updates

For the next calculation let $AX_i \sim \mathcal{N}(a, V)$ and $BX_j \sim \mathcal{N}(b_j, W_j)$ be independent, with $V, W_j \succ 0$, and take $o = 0$. The likelihood score and the direct peer score are

$$C_j = c_R + \tfrac{1}{2}\left[(a-b_j)^T R^{-1}(a-b_j) + \operatorname{tr}(R^{-1}(V+W_j))\right], \tag{S3.3}$$

$$D_j = \tfrac{1}{2}\left[(a-b_j)^T W_j^{-1}(a-b_j) + \operatorname{tr}(W_j^{-1}V) - d + \log\frac{\det W_j}{\det V}\right]. \tag{S3.4}$$

The second formula follows by taking the expectation of the log ratio of the two Gaussian densities. Its receiver quadratic has expectation $d$, while its source quadratic has expectation $\operatorname{tr}(W_j^{-1}V) + (a-b_j)^T W_j^{-1}(a-b_j)$.

**Proposition S3.2 (A restricted routing equivalence).** Suppose $W_j = W$ for every source, $W$ is held fixed, and $R = TW$ for $T > 0$. Then

$$C_j - \frac{D_j}{T} = c_{TW} + \frac{d}{T} - \frac{1}{2T}\log\frac{\det W}{\det V}. \tag{S3.5}$$

Consequently normalized rows proportional to $\pi_j e^{-C_j}$ and to $\pi_j e^{-D_j/T}$ coincide. Their derivatives with respect to receiver covariance nevertheless differ by

$$d_V C_j - d_V(D_j/T) = \frac{1}{2T}\operatorname{tr}(V^{-1}dV). \tag{S3.6}$$

*Proof.* Insert $R = TW$ and $W_j = W$ into eqs. (S3.3) and (S3.4). The mean quadratic and $\operatorname{tr}(W^{-1}V)$ cancel in their difference, leaving eq. (S3.5), which is independent of $j$. A common additive score cancels on row normalization. Differentiating the remaining $\log\det V$ gives eq. (S3.6). □

At $T = 1$ the score difference is $H(\mathcal{N}(a, V)) + d/2$, with entropy measured against Lebesgue measure. It is constant for source selection, but not for variation of the receiver law. If $V$ is also fixed, the corresponding mean derivatives agree. If source covariances vary independently, or the comparison maps change them, the hypothesis of a common held $W$ no longer applies. These are restrictions on the optimization block, not alternative formulas for one universal interaction.

## S3.3 Identity self-comparisons and independent snapshots

Take $A = B = I_d$ and let a self comparison use the same random variable twice. Then $X_i - X_i = 0$ identically, so

$$\mathbb{E}[-\log p(o \mid X_i, X_i)] = c_R + \tfrac{1}{2}o^T R^{-1} o. \tag{S3.7}$$

Its derivative with respect to the law of $X_i$ is zero. In eq. (S3.2), all four covariance terms cancel because $\Sigma_{ii}$ is also the cross covariance. The zero-record identity peer divergence $D_{\mathrm{KL}}(q_i \| q_i) = 0$ likewise has zero total derivative.

Now let $X_i$ and $\widetilde{X}_i$ be independent variables with the same $\mathcal{N}(0, I_d)$ law, and compare their difference with noise $R = TI_d$. At zero record their cost is $c_R + d/T$. Thus an identity self candidate and a distinct independent candidate with the same marginal law receive different likelihood scores,

while their peer divergences are both zero. With equal source priors, the likelihood odds in favor of the identity self candidate are $e^{d/T}$.

An independently sampled self snapshot realizes the second calculation. A remembered state can instead be correlated with the current state, in which case its joint law determines an intermediate or different cost through eq. (S3.2). Calling a candidate "self" does not select that joint law. The distinction must be fixed before computing a comparison or a receiving update.

## S3.4 Integrating the source before taking a logarithm

Let $X \sim \mathcal{N}(a, V)$, $Y \sim \mathcal{N}(b, W)$ be independent, with $V \succ 0$, $W \succeq 0$, and $R \succ 0$. Regard $k(x \mid y) = \mathcal{N}(x; y, R)$ as a normalized receiver-coordinate kernel. Integrating over the sender gives $r(x) = \mathcal{N}(x; b, W + R)$. Therefore

$$-\mathbb{E}_X \log r(X) = c_{W+R} + \tfrac{1}{2}\left[(a-b)^T (W+R)^{-1}(a-b) + \operatorname{tr}((W+R)^{-1}V)\right]. \tag{S3.8}$$

This expression compares the receiver to a noise-convolved source. It is not the peer divergence against $\mathcal{N}(b, W)$, even when that law is nonsingular.

Subtracting eq. (S3.8) from the expected negative log kernel gives the exact Jensen gap

$$\begin{aligned} G = {} & \tfrac{1}{2}\log\frac{\det R}{\det(W+R)} + \tfrac{1}{2}\operatorname{tr}(R^{-1}W) \\ & + \tfrac{1}{2}(a-b)^T\left[R^{-1} - (W+R)^{-1}\right](a-b) \\ & + \tfrac{1}{2}\operatorname{tr}\left(\left[R^{-1} - (W+R)^{-1}\right]V\right). \end{aligned} \tag{S3.9}$$

For completeness, $R^{-1} - (W+R)^{-1}$ is positive semidefinite: conjugating by $R^{1/2}$ reduces this assertion to $I - (I + R^{-1/2} W R^{-1/2})^{-1} \succeq 0$. The first line of eq. (S3.9) is half the sum of $\lambda - \log(1+\lambda)$ over the nonnegative eigenvalues of $R^{-1/2} W R^{-1/2}$. Each summand is nonnegative because its derivative is $\lambda/(1+\lambda)$ and its value at zero is zero. Hence $G \geq 0$, with $G > 0$ whenever $W \neq 0$. Averaging the conditional log density and taking the log of the integrated density remain different even at coincident means.

# Chapter S4

# Supporting Constructions for Inheritance

This chapter gives two exact ways to determine what a parent must retain: factorization of a boundary map and conditional prediction from a retained history. Both distinguish an inherited response from a newly chosen parent interaction.

## S4.1 Linear boundary descent

**Proposition S4.1 (Descent through a surjective retention).** Let $E_j, E_i, Z_j, Z_i$ be finite-dimensional real vector spaces. Let $C_j : E_j \to Z_j$ be surjective, let $C_i : E_i \to Z_i$ be linear, and let $M_{ij} : E_j \to E_i$ be a linear boundary map. There is a linear map $M^c_{IJ} : Z_j \to Z_i$ satisfying

$$C_i M_{ij} = M^c_{IJ} C_j, \tag{S4.1}$$

if and only if

$$\ker C_j \subseteq \ker(C_i M_{ij}). \tag{S4.2}$$

When it exists, it is unique.

*Proof.* If (S4.1) holds, every $v \in \ker C_j$ satisfies $C_i M_{ij} v = 0$. Conversely, define $M^c_{IJ}(C_j v) = C_i M_{ij} v$. Two choices $v, w$ with $C_j v = C_j w$ differ by an element of $\ker C_j$, which (S4.2) annihilates. The definition is therefore independent of the representative. It is linear, and surjectivity of $C_j$ defines it everywhere. The same surjectivity proves uniqueness. □

For example, let $C_j(u, v) = u$ and $C_i$ be the identity on $\mathbb{R}$. The map $M_{ij}(u, v) = u + v$ cannot descend: the discarded vector $(0, 1)$ changes the retained output. The map $M_{ij}(u, v) = 2u$ descends to multiplication by two. Agreement of two current input laws does not change either conclusion, since the requirement concerns the boundary mechanism on its declared domain.

The same criterion handles an additive-noise channel without requiring Gaussian noise. Suppose the receiver content is $M_{ij} v + \varepsilon$, where $\varepsilon$ has a fixed probability law $\nu$ independent of $v$. If (S4.2) holds, its retained law is the distribution of

$$M^c_{IJ} C_j v + C_i \varepsilon. \tag{S4.3}$$

Thus the parent inherits the pushed noise law $(C_i)_{\#}\nu$. A state-dependent noise law requires its own factorization through $C_j$; factoring the mean map alone is insufficient.

## S4.2 Exact finite filtering

Let $X_n$ be a Markov chain on a finite configuration space S with transition matrix $T$ and initial row distribution $p_0$. Let $Z_n = R(X_n)$ for a map into a finite retained space $\mathsf{S}^c$. For each $z$, let $P_z$ be the diagonal matrix whose entry at $x$ is $1_{\{R(x)=z\}}$. These matrices are selectors of configurations, not new dynamical transitions.

**Proposition S4.2 (The terminal conditional law retains every future prediction).** For a retained history $h_n = (z_0, \dots, z_n)$ of positive probability, define

$$f_{h_n} = p_0 P_{z_0} T P_{z_1} \cdots T P_{z_n}, \qquad \widehat{p}_{h_n} = \frac{f_{h_n}}{f_{h_n}\mathbf{1}}. \tag{S4.4}$$

Then $\widehat{p}_{h_n}$ is the conditional distribution of $X_n$ given $h_n$, and

$$\Pr(Z_{n+1} = z \mid h_n) = \widehat{p}_{h_n} T P_z \mathbf{1}. \tag{S4.5}$$

After a positive-probability next value $z$, the update is

$$\widehat{p}_{(h_n,z)} = \frac{\widehat{p}_{h_n} T P_z}{\widehat{p}_{h_n} T P_z \mathbf{1}}. \tag{S4.6}$$

For any finite future word, its probability is obtained by further multiplication by the corresponding $TP_z$ matrices and by $\mathbf{1}$.

*Proof.* The $x$ component of $p_0 P_{z_0}$ is the joint probability of $X_0 = x$ and $Z_0 = z_0$. Multiplication by $T$ propagates this joint mass one step; multiplication by $P_{z_1}$ restricts it to the next retained value. Induction gives the joint mass of $h_n$ and $X_n = x$ as the $x$ component of $f_{h_n}$. Its sum is the probability of $h_n$, so division gives the conditional distribution. One further propagation and selection gives (S4.5) and (S4.6). Repeating the operation gives every future word probability. □

A zero-probability history has zero unnormalized message and no conditional law supplied by this formula. Because all matrices are nonnegative, every extension of that history also has zero mass. The construction therefore needs no arbitrary prediction on an impossible history.

The filter is sufficient even when the retained process is not Markov. For the deterministic cycle $1 \to 2 \to 3 \to 1$ with uniform initial law, set $R(1) = R(2) = 0$ and $R(3) = 1$. The histories $(0, 0)$ and $(1, 0)$ both have positive probability and end at the same retained value. The first ends at state $2$ and predicts $1$ next; the second ends at state $1$ and predicts $0$. Hence the current retained value cannot determine its next distribution. The terminal conditional law does.

## S4.3 Successive retention of responses

Suppose a declared joint law contains a fine state $X$ and a boundary response $O$. Let $Z = R_{10}(X)$ and $W = R_{21}(Z)$, and assume all spaces are standard Borel. For every bounded measurable response observable $g$, conditional expectation gives

$$\mathbb{E}[g(O) \mid W] = \mathbb{E}\big[\mathbb{E}[g(O) \mid Z] \mid W\big]. \tag{S4.7}$$

To verify the identity, integrate either side over any event determined by $W$. Such an event is also determined by $Z$, so both integrals equal the integral of $g(O)$ over that event. This characterizes conditional expectation up to null sets.

The identity concerns one declared joint law. Applying it to an equilibrium law produces equilibrium-conditional responses. Applying it to the joint law conditional on a retained history produces the corresponding predictive responses. These are interchangeable only when their relevant conditional laws agree. Exact boundary descent is stronger: it fixes the response on every admitted fiber, independently of the conditional weights used there.

# Bibliography

[1] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.